SUPPORTING ADULT GUIDANCE

Platonic Solids

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner meets the five Platonic solids, verifies Euler's formula F + V – E = 2 across them, and reasons about why there are exactly five — from the condition that the angles meeting at a corner must be less than 360°. The emphasis is on **a complete piece of reasoning**: seeing why no sixth regular solid can exist.

Driving Question: *Why are there exactly five perfectly regular solids — no more, no fewer?*

The mathematics, in plain terms

A **Platonic solid** is a convex 3D shape where every face is the same regular polygon and the same number of faces meet at each corner. There are exactly five: the tetrahedron (4 triangles), the cube (6 squares), the octahedron (8 triangles), the dodecahedron (12 pentagons) and the icosahedron (20 triangles). For every one — indeed for every convex polyhedron — **F + V – E = 2**, where F, V and E are the numbers of faces, vertices and edges; this is Euler's formula. Why only five? At a corner, at least three faces must meet, and their angles must add up to **less than 360°** to leave a gap that folds into 3D. Equilateral triangles (60°) allow three, four or five at a corner; squares (90°) allow three; regular pentagons (108°) allow three. Six triangles, four squares or three hexagons make exactly 360° and lie flat, and anything more overlaps — so no sixth solid is possible. The solids also form **dual pairs**: joining the centres of one solid's faces gives another (cube and octahedron, dodecahedron and icosahedron, and the tetrahedron with itself).

A gap at every corner

The heart of this lesson is one beautiful argument: a regular solid can exist only if the identical faces meeting at each corner leave a gap — their angles totalling less than 360°. The most useful thing to protect is that the learner reaches this idea themselves, by trying each polygon in turn and watching the gap open, close or overlap. It builds directly on the angles lesson: the same corner-fitting that decided which shapes tile a flat surface now decides which fold into a solid. Euler's F + V – E = 2 is a second thread — do not push the learner to memorise it, but let them notice that it holds every single time.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The film about Euler, then verifying F + V – E = 2 across all five solids — the invariant that ties them together.	~15 min

Movement	Folding each solid up from its flat net and rotating it in 3D to see every face.	~8 min
Reflection	The Corner Investigator: reasoning about why only five combinations of angles leave a gap and fold into a solid.	~10 min
Creativity	Identifying solids from their properties, and discovering the dual hidden inside each one.	~5 min
Rest	A pause among five shapes that have fascinated people for thousands of years.	~4 min
Nutrition	Contextual (Mode A): fair dice — the d4, d6, d8, d12 and d20 are exactly the five Platonic solids.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking there could be a sixth Platonic solid

Repair: “What is the smallest regular polygon that cannot work at a corner, and why? What do three regular hexagons add up to?” (3 × 120° = 360°, so no gap.)

Confusing 'regular' with any symmetric polyhedron

A shape can be 3D and symmetric without being Platonic. Repair: “Are all its faces the same regular polygon, and do the same number meet at every corner?”

Muddling faces, edges and vertices in Euler's formula

Repair: “Point to a face, an edge, a vertex. Now count each carefully — including the ones hidden at the back.”

Assuming the angles at a corner must add to 360°

That is the condition for a flat tiling. Repair: “To fold up into a corner, do the angles need to fill the point, or to leave a gap?”

Thinking a dual is just a smaller copy

Repair: “Count the faces of the first solid — that is how many vertices the dual has. Is the dual the same shape, or a different one?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the faces, edges and vertices of a solid, the interior angle of a regular polygon, and angles around a point. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the reasoning — and the corner — with the learner:

- “How many faces meet at one corner of this solid? What do their angles add up to?”
- “Is that under 360°? So does it leave a gap to fold up into a corner?”
- “Count F, V and E — does F + V – E come to 2?”
- “If you joined the centres of this solid's faces, which solid would you get?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise all five and their numbers?”

Recognising the five and being able to reason about them matters far more than rote recall. The Movement and Corner tools let a learner rebuild the facts rather than memorise them.

“Why does the ‘less than 360°’ idea matter so much?”

It is the whole proof that there are exactly five — a rare chance for a learner to see a complete, watertight argument, and it grows straight out of the angles lesson.

“They find the dual idea hard to picture.”

Use the on-screen dual viewer, which puts a dot at the centre of each face and joins them, and the isometric grid to sketch. Handling a real dice — a d6 is a cube, a d20 an icosahedron — makes it concrete.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.